

AirNav Risk Analysis Platform (ARAP)

Requirements for an application to conduct safety risk assessments with the twelve methods of the AirNav Risk Analysis Manual

Audience: software developers, architects, testers and the AirNav product owner

Structure follows ISO/IEC/IEEE 29148:2018; quality attributes follow ISO/IEC 25010

Companion to: AirNav Risk Analysis Manual, Edition 0.1

Document control

Item	Detail
Document	Software Requirements Specification — AirNav Risk Analysis Platform (ARAP)
Version / status	0.1 — draft for review by the product owner, Safety Manager and IT architecture board
Product owner	Safety Manager, AirNav (to be confirmed)
Source of domain requirements	AirNav Risk Analysis Manual, Ed. 0.1 (the “Manual”). Where this SRS and the Manual differ on method content, the Manual prevails and this SRS shall be corrected.

Conventions

Requirements use **shall** (mandatory), **should** (recommended) and **may** (optional). Each has a unique identifier, a priority using MoSCoW — **M** must (release 1 blocker), **S** should, **C** could — and a verification method: **T** test, **D** demonstration, **I** inspection, **A** analysis. Identifiers are stable: a deleted requirement keeps its number and is marked “withdrawn”.

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1. Introduction

1.1 Purpose

This document specifies the requirements for the AirNav Risk Analysis Platform (ARAP), a web application that lets AirNav staff plan, conduct, review, approve and maintain safety risk assessments using the twelve methods described in the Manual: Bowtie, HAZID, HAZOP, JHA, FMEA/FMECA, LOPA, FHA, STPA, FTA, bio-mathematical fatigue modelling (three-process model), FRAM and Bayesian Belief Networks. It is written for the development team (in-house or contracted) and for those who will test and accept the product.

1.2 Scope

ARAP replaces spreadsheets and stand-alone drawing tools with one system that holds a single hazard log, applies one risk classification scheme, performs the calculations the methods need, keeps a complete audit trail and generates the Safety Assessment Report. In scope: assessment management, all twelve method modules, calculation engines, hazard log, actions and approval workflow, reporting, libraries, administration and integrations listed in Section 7. Out of scope: occurrence reporting and investigation case management (integrated, not replaced), roster production (integrated), and any function used directly for live air traffic control.

1.3 Product goals

- **Consistency:** every assessment follows the process and risk classification scheme of the Manual, whatever method is used.
- **Traceability:** every hazard, control, requirement and action can be traced to its source analysis, its evidence and its approval.
- **Correctness:** quantitative results are computed by verified, versioned engines that reproduce the Manual's worked examples (Section 9).
- **Living risk picture:** results stay connected to safety performance indicators and occurrence data after approval.
- **Usability:** a facilitator can run a live workshop in the tool, projecting worksheets and diagrams, without a separate scribe tool.

1.4 References

Ref.	Document
[M]	AirNav Risk Analysis Manual, Edition 0.1 (draft), September 2026 — method definitions, worksheets, worked examples.
[S1]	ISO/IEC/IEEE 29148:2018 Systems and software engineering — Life cycle processes — Requirements engineering.
[S2]	ISO/IEC 25010 Systems and software Quality Requirements and Evaluation (SQuaRE) — product quality model.
[S3]	ISO/IEC 27001:2022 Information security management systems; OWASP Application Security Verification Standard (ASVS).
[S4]	Undang-Undang Nomor 27 Tahun 2022 tentang Pelindungan Data Pribadi (Indonesian Personal Data Protection Law).
[S5]	W3C Web Content Accessibility Guidelines (WCAG) 2.2.
[S6]	ICAO Annex 19, ICAO Doc 9859 (4th ed.), CASR Part 19 (PM 2 Tahun 2026) — regulatory basis of the SMS processes supported.

2. Overall description

2.1 Product perspective

ARAP is a new, self-contained web application with a service-oriented back end. It exchanges data with AirNav's identity provider, occurrence reporting system, rostering system, maintenance management system and document management system (Section 7). Figure 1 shows the logical architecture; it is indicative, and the supplier may propose an equivalent design that meets the requirements.

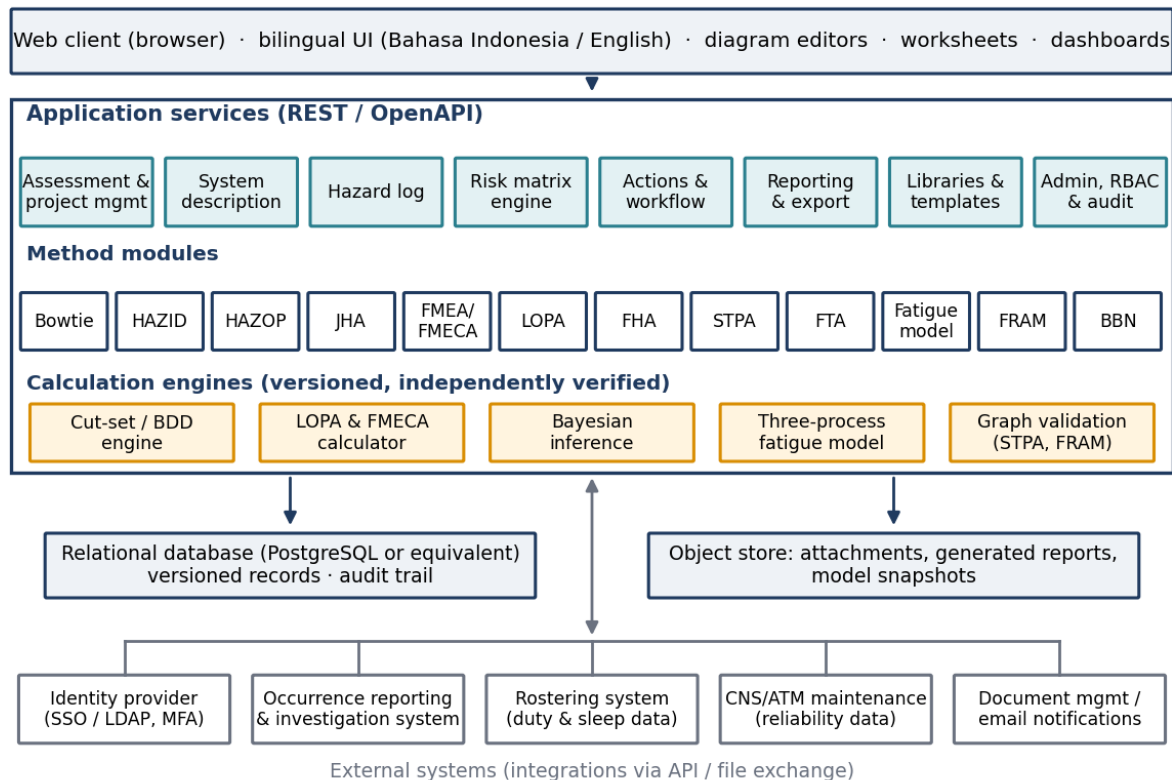


Figure 1. Indicative logical architecture of ARAP.

2.2 User roles

Role	Typical user	Main permissions
Viewer	Any authorised staff member	Read approved assessments, bowties and the hazard log within their organisational scope.
Participant	SME invited to a study	Read the study; comment; contribute to worksheets while the session is open.
Assessor / facilitator	Safety officer, trained facilitator	Create and edit assessments and studies; run sessions; propose risk ratings; submit for review.
Reviewer	Safety Manager or delegate	Review, return with comments, or endorse; edit libraries and templates.
Risk acceptance authority	Unit manager, Director, Accountable Executive	Accept or reject residual risk within their delegated authority (Manual Table 3.3).
Action owner	Any staff member assigned an action	Update status and evidence of their actions.
Fatigue analyst	FRMS specialist	Run fatigue models; access identifiable roster/sleep data (restricted).
Administrator	IT / SQRM admin	Users, roles, organisational structure, risk scheme configuration, integrations. No authority to approve assessments.

2.3 Operating environment

- Browser-based client supporting the two most recent major versions of Chrome, Edge, Firefox and Safari; usable on a 1366 × 768 laptop and projected at 1920 × 1080; read-only views usable on tablets.
- Server deployment on AirNav infrastructure or an approved cloud hosted in Indonesia; containerised deployment (e.g. Docker/Kubernetes) preferred.
- Offline or low-bandwidth use at remote units is supported for data entry by a workshop “offline pack” (Section 4.1, COM-15).

2.4 Constraints and assumptions

- ARAP is a safety-management support tool. It is not used for live ATC and is not subject to ATM software assurance levels; nevertheless its calculation engines shall be verified (Section 6.6) because their results support safety decisions.
- The risk classification scheme, likelihood bands, tolerability regions and acceptance authorities are configuration data, not code (Manual Chapter 3 values are the initial configuration).
- All user interface text shall be available in Bahasa Indonesia and English; stored analysis content is kept in the language in which it was entered.
- Fatigue and roster data are personal data and are processed under [S4].

3. Domain model

The core concept is that every method produces the same kinds of objects — hazards, causes, consequences, controls, risk assessments, requirements and actions — which are stored once and shared. Method-specific objects (a HAZOP node, a fault-tree gate, a BBN node) are linked to these common objects. Figure 2 shows the principal entities.

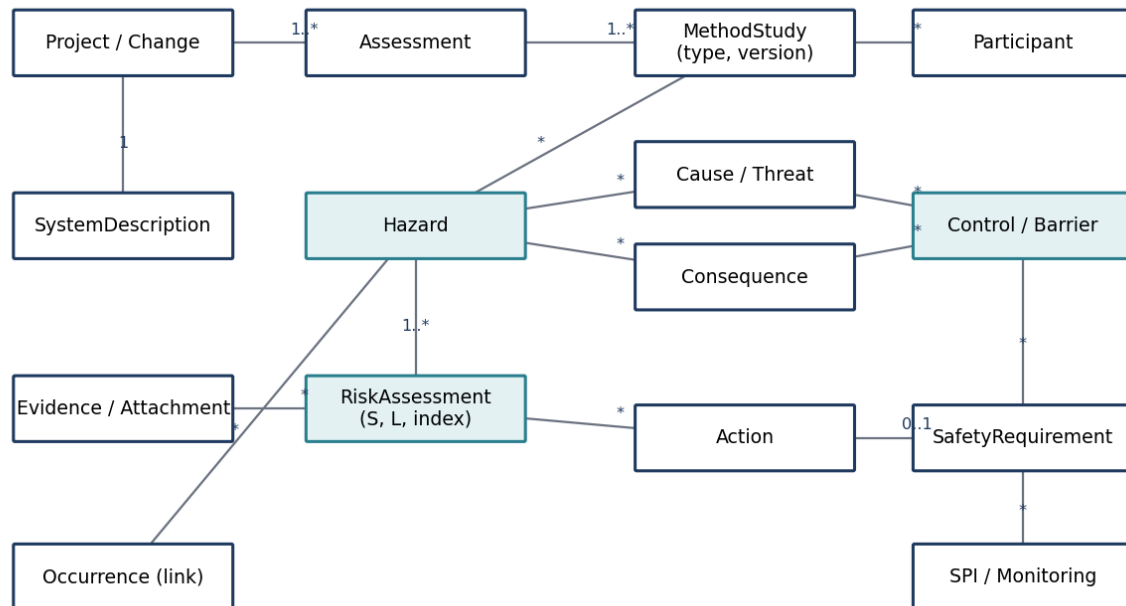


Figure 2. Principal entities and relationships (method-specific entities omitted).

Entity	Key attributes (minimum)
Project / Change	ID, title, type (system, procedure, airspace, organisational, temporary), units affected, sponsor, dates, status
Assessment	ID, project, scope, assumptions, status (draft, in review, endorsed, accepted, closed, superseded), version, approvals
MethodStudy	ID, assessment, method type, method template version, sessions (date, participants, duration), status
Hazard	Global hazard-log ID, title, description, operational context, source studies, owner, status, review date
Cause / Threat; Consequence	ID, text, category (People, Procedures, Equipment, Environment, ...), linked hazard(s)
Control / Barrier	ID, text, type (hardware, software, human, human-hardware, procedure, organisational), side (prevention/recovery), owner, verification status (existing-verified, existing-unverified, planned), effectiveness, criticality, linked SPIs
RiskAssessment	Hazard, consequence, stage (initial, current, residual, target), severity, likelihood, index, region, rationale, evidence links, assessor, timestamp
SafetyRequirement	ID, text, source (FHA objective, LOPA gap, STPA constraint, ...), verification method, status, evidence
Action	ID, text, owner, due date, status, closure evidence, linked hazard/control/requirement
Evidence	File or link, description, source, date, confidentiality marking

4. Functional requirements

4.1 Common functions (COM)

ID	Requirement	Pri.	Verif.
COM-01	The system shall allow an assessor to create a project/change and one or more assessments under it, recording scope, boundary, operational environment, assumptions and before/after description (Manual §2.3).	M	T
COM-02	The system shall allow one or more method studies of any of the twelve method types to be added to an assessment, and shall record which template version each study used.	M	T
COM-03	The system shall recommend candidate methods from the change type and lifecycle phase using the Manual's selection guidance (Manual Table 4.1 and §4.1); the recommendation shall be advisory only.	S	D
COM-04	The system shall provide a session mode for live workshops: participant list and attendance, timer, presentation layout for projection, and capture of dissenting views against any row.	M	D
COM-05	Every hazard, cause, consequence and control created in any method study shall be stored as, or linked to, a common object in the hazard log, with the source study and row recorded.	M	T
COM-06	The system shall detect probable duplicate hazards and controls (text similarity and shared links) and offer to merge them, keeping all source links.	S	T
COM-07	Risk assessments shall use the configured risk classification scheme; the user selects severity and likelihood and the system derives the risk index, tolerability region and required acceptance authority.	M	T
COM-08	The system shall require a rationale and allow evidence links for every severity and likelihood judgement; it shall flag assessments where no evidence is linked.	M	T
COM-09	Only controls with status "existing-verified" shall be credited in the current-risk assessment; planned controls may only be credited in the residual (target) risk.	M	T
COM-10	Where a quantitative method (FTA, LOPA, BBN, FMECA, FHA objectives) produces a frequency or probability, the system shall map it to a likelihood level using the configured quantitative bands and show the unit of measure; comparing values with different units shall be blocked.	M	T
COM-11	The system shall support the workflow Draft → In review → Endorsed → Accepted / Rejected → Closed, with configurable reviewers and acceptance authority derived from the highest residual risk region in the assessment.	M	T
COM-12	Actions shall be assignable to any user with due date and reminders (e-mail and in-app); overdue actions shall be escalated to the assessment owner and the Safety Manager.	M	T
COM-13	Endorsed or accepted assessments shall be locked; changes require a new version that shows a comparison with the previous version.	M	T
COM-14	Comments shall be possible on any object, with mention of users and resolution status.	S	D
COM-15	An offline workshop pack (exported worksheet file) should allow data capture without network access and later re-import with conflict detection.	C	T
COM-16	Global search shall find hazards, controls, requirements, actions and studies by ID, text, unit, system, method and status.	M	T

4.2 Hazard log (HZL)

ID	Requirement	Pri.	Verif.
HZL-01	The hazard log shall hold every hazard with the minimum content listed in Manual §18.2 and a unique, never-reused identifier.	M	I
HZL-02	The hazard log shall show, for each hazard, all linked studies, controls, risk assessments (initial, current, residual), requirements, actions, SPIs and occurrences.	M	D
HZL-03	Users shall be able to filter and export the hazard log (XLSX, CSV) by unit, system, risk region, owner, status and review date.	M	T
HZL-04	The system shall notify hazard owners 30 days before a review date and flag overdue reviews on the dashboard.	M	T
HZL-05	Linking an occurrence (from the occurrence reporting system) to a hazard and to the barrier(s) that failed shall be possible, and barrier-failure counts shall feed the bowtie barrier view.	S	T

4.3 Bowtie module (BOW)

ID	Requirement	Pri.	Verif.
BOW-01	The module shall provide a graphical bowtie editor with hazard, top event, threats, consequences, prevention barriers, recovery barriers, escalation factors and escalation-factor barriers (Manual §6.1).	M	D
BOW-02	Barriers shall carry attributes: type, owner, effectiveness (configurable scale, default very good / good / poor / very poor / unknown), criticality, verification status, linked SPIs/audit items and free text.	M	T
BOW-03	The same barrier object may appear on several paths; editing it in one place shall update all occurrences and the diagram shall indicate shared barriers.	M	T
BOW-04	The editor shall automatically lay out the diagram and allow manual adjustment; it shall support collapse/expand of paths and colour display by effectiveness or owner.	M	D
BOW-05	The module shall generate the barrier register table (Manual §6.3) and export the diagram as SVG, PNG and PDF.	M	T
BOW-06	A bowtie library shall hold approved template bowties (e.g. those adapted from the UK CAA templates, subject to their terms of use) that can be copied into an assessment and adapted.	S	D
BOW-07	The module should import bowtie models from a documented JSON schema and from CSV, and should attempt import of BowTieXP XML exports.	S	T
BOW-08	The module shall warn when a consequence path has fewer than two barriers, when a barrier has no owner, or when a top event is phrased as a consequence (keyword check, advisory).	S	T

4.4 HAZID module (HZD)

ID	Requirement	Pri.	Verif.
HZD-01	The module shall provide a HAZID register with columns: ID, guideword/category, hazard, causes, consequences, existing controls, severity, likelihood, risk index, actions, owner (Manual §7.4).	M	T
HZD-02	Guideword checklists shall be configurable libraries; the ANSP checklist of Manual Table 7.2 shall be supplied as default.	M	I
HZD-03	The facilitator shall be able to step through guidewords one at a time in session mode and record “no hazard identified” for a guideword.	M	D
HZD-04	Any row may be flagged “needs detailed method” with the proposed method; the flag shall create a draft study of that type linked to the hazard.	S	T

4.5 HAZOP module (HZIP)

ID	Requirement	Pri.	Verif.
HZIP-01	The module shall allow the design to be divided into nodes, each with a design intent, drawings/attachments and parameters.	M	T
HZIP-02	The module shall generate the deviation matrix from selected parameters × guidewords (default set: No/Not, More, Less, As well as, Part of, Reverse, Other than, Early, Late, Before, After — Manual Table 8.1) and let the team mark each deviation as not meaningful, or analyse it.	M	T
HZIP-03	For each analysed deviation the worksheet shall capture causes, consequences, safeguards, risk index and recommendations, and show node progress (deviations analysed / total).	M	T
HZIP-04	The module shall support procedural HAZOP (steps of a procedure as nodes) and data-flow HAZOP (messages as nodes) with separate default parameter sets.	S	D

4.6 JHA module (JHA)

ID	Requirement	Pri.	Verif.
JHA-01	The module shall record a job (title, location, equipment, frequency, required permits) and its ordered steps, with hazards and controls per step (Manual §9.3).	M	T
JHA-02	Each control shall be classified by the hierarchy of controls (elimination, substitution, engineering, administrative, PPE); the module shall warn when all controls on a step are administrative or PPE.	M	T
JHA-03	A JHA shall be printable as a one-page (A4 landscape) briefing sheet and linkable to work orders in the maintenance system.	S	D
JHA-04	Where a job affects an operational service (e.g. equipment out of service), the module shall prompt for the ATS coordination required and link to the relevant hazard.	S	D

4.7 FMEA / FMECA module (FME)

ID	Requirement	Pri.	Verif.
FME-01	The module shall hold an item hierarchy (system → subsystem → item) importable from CSV/XLSX, with failure rate and data source per item.	M	T
FME-02	The worksheet shall capture, per failure mode: cause, local effect, end effect on ATS, detection method, compensating provisions, severity, occurrence and detection ratings (Manual §10.3).	M	T
FME-03	The module shall compute $RPN = S \times O \times D$ using configurable 1–10 scales that are displayed with their definitions, and shall highlight high-severity modes regardless of RPN.	M	T
FME-04	The module shall compute MIL-STD-1629A failure-mode criticality $C_m = \beta \cdot \alpha \cdot \lambda_p \cdot t$ and item criticality per severity class, with α per item summing to 1 (validation error otherwise).	S	T
FME-05	The module shall list single points of failure and allow export of failure modes as basic events into the FTA module.	S	T

4.8 LOPA module (LOP)

ID	Requirement	Pri.	Verif.
LOP-01	The module shall record scenarios with initiating event and frequency (with unit and data source), conditional modifiers/enabling conditions, candidate safeguards and a tolerable frequency target.	M	T
LOP-02	For each safeguard the user shall answer the four IPL criteria (independent, effective, dependable, auditable) with justification; only safeguards meeting all four may be credited, with a PFD.	M	T
LOP-03	The module shall compute $f_C = f_{IE} \times \Pi P_{CM} \times \Pi PFD$, the ratio to target and the additional risk-reduction factor needed, and chart the reduction per layer on a log scale (Manual Figure 5).	M	T
LOP-04	The module shall warn when two credited IPLs share a declared dependency (e.g. same surveillance source, same person) and require an explicit override justification.	S	T
LOP-05	Libraries of initiating-event frequencies and generic PFDs shall be maintainable by reviewers, each value with source and validity date.	S	I

4.9 FHA module (FHA)

ID	Requirement	Pri.	Verif.
FHA-01	The module shall record system functions and, for each, failure conditions of the types total loss, partial loss/degradation, erroneous (detected / undetected), unintended and delayed (Manual §12.1).	M	T
FHA-02	For each failure condition the module shall capture operational effects, context assumptions, severity and the derived safety objective, computed from the configured scheme as the maximum frequency that keeps the risk at or below the configured target region.	M	T
FHA-03	The module shall allow apportionment of an objective among failure conditions contributing to the same outcome, with a check that the apportioned sum does not exceed the objective.	S	T
FHA-04	Safety objectives and derived requirements shall be exportable to the FTA module as top events with targets, and to the requirement register.	M	T

4.10 STPA module (STP)

ID	Requirement	Pri.	Verif.
STP-01	The module shall record losses, system-level hazards (linked to losses) and system-level constraints (linked to hazards) (Manual §13.1 step 1).	M	T
STP-02	A control-structure editor shall allow controllers, controlled processes, control actions and feedback paths to be drawn hierarchically, and each controller to hold a process-model description.	M	D
STP-03	For every control action the module shall generate a UCA table with the four types (not providing; providing; too early/too late/wrong order; stopped too soon/applied too long), allowing “not applicable” with reason.	M	T
STP-04	Each UCA shall require a context (“when ...”) and at least one linked hazard; the module shall report UCAs without context or hazard links.	M	T
STP-05	The module shall record loss scenarios per UCA (types: inadequate feedback, flawed process model, conflicting control, component failure, control action not executed properly) and derive requirements/constraints from them.	M	T
STP-06	A traceability view shall show loss → hazard → constraint → UCA → scenario → requirement, and flag broken chains.	S	D
STP-07	The module should support CAST (accident/incident analysis) reusing the same control structure.	C	D

4.11 FTA module (FTA)

ID	Requirement	Pri.	Verif.
FTA-01	The module shall provide a fault-tree editor supporting top event, intermediate events, basic, undeveloped and house events, and OR, AND, k-out-of-n (voting) and NOT-free logic; transfer gates shall allow a subtree to be reused.	M	D
FTA-02	Basic events shall accept a probability, or a failure rate with exposure time, or an unavailability model (λ , MTTR or test interval), and shall convert them to probability for the chosen mission time, displaying the formula used.	M	T
FTA-03	The engine shall compute minimal cut sets exactly (e.g. MOCUS or binary decision diagrams) for trees up to 2 000 basic events, with an order/probability truncation option for larger trees, and report truncation.	M	T
FTA-04	The engine shall compute the top-event probability both exactly (BDD) and by the rare-event approximation, and display both when they differ by more than 1%.	M	T
FTA-05	The engine shall compute Birnbaum and Fussell-Vesely importance for every basic event and rank cut sets by contribution.	M	T
FTA-06	Common-cause failure groups shall be definable with the β -factor model, and the engine shall add the CCF events to the cut sets automatically.	S	T
FTA-07	The module shall compare the result with a linked FHA objective and show pass/fail and margin; sensitivity analysis shall vary any input over a user-defined range.	S	T
FTA-08	Fault trees shall be exportable as SVG/PDF and in the Open-PSA Model Exchange Format.	S	T

4.12 Fatigue modelling module (FAT)

ID	Requirement	Pri.	Verif.
FAT-01	The module shall import duty schedules from the rostering system (API or CSV) and allow manual entry of rosters and roster templates.	M	T
FAT-02	Sleep periods shall be generated by configurable rules (e.g. normal night sleep window, day sleep after night duty, optional naps) or imported from sleep diaries/actigraphy; generated and measured sleep shall be distinguished.	M	T
FAT-03	The engine shall implement the three-process model as specified in Section 5.1, with all parameters configurable and the default values of the Manual (Table in §15.2).	M	T
FAT-04	For each duty the module shall report minimum alertness and its clock time, maximum KSS, time and share of duty at or above a configurable KSS threshold (default 7), and plot alertness over time.	M	T
FAT-05	Users shall be able to compare two or more roster variants side by side and export the comparison.	M	D

ID	Requirement	Pri.	Verif.
FAT-06	Aggregated, anonymised roster-level results shall be the default view; identifiable individual results shall be visible only to the Fatigue analyst role and logged on access.	M	T
FAT-07	The module should allow calibration of parameters against collected KSS data (e.g. fitting the KSS intercept and slope) and store calibrated parameter sets with version and date.	C	T
FAT-08	Outputs shall carry the notice that model predictions are group averages and not a measure of an individual's fatigue.	M	I

4.13 FRAM module (FRM)

ID	Requirement	Pri.	Verif.
FRM-01	The module shall provide a FRAM editor with hexagonal functions and the six aspects (Input, Output, Precondition, Resource, Control, Time), each aspect holding free-text entries.	M	D
FRM-02	Couplings shall be created by matching an Output text to an aspect of another function; the module shall list unmatched aspects ("orphans") for validation.	M	T
FRM-03	Each function shall have a type (human, technological, organisational) and, per instantiation, output variability for timing (too early, on time, too late, not at all) and precision (precise, acceptable, imprecise).	M	T
FRM-04	The module shall support instantiations (scenarios) that highlight the active couplings and propagate variability downstream for review.	S	D
FRM-05	Models should be importable and exportable in the FRAM Model Visualiser file format.	S	T

4.14 BBN module (BBN)

ID	Requirement	Pri.	Verif.
BBN-01	The module shall provide a graph editor for discrete-state nodes and directed arcs and reject cycles.	M	T
BBN-02	A CPT editor shall enforce that each column sums to 1 (within 1e-9), support copy/paste from spreadsheets, and record the source of each table (data, expert elicitation with names/roles, literature).	M	T
BBN-03	The engine shall perform exact inference (variable elimination or junction tree) for networks up to 60 nodes with up to 5 states and 4 parents per node, returning marginals within 2 s.	M	T
BBN-04	Users shall be able to set hard evidence on any node and see updated marginals (predictive and diagnostic queries), and save named scenarios.	M	T
BBN-05	The module shall provide sensitivity analysis (tornado chart) of a target node to each CPT parameter or parent node.	S	T
BBN-06	Parametric CPT helpers (noisy-OR, ranked nodes) should be provided.	C	T
BBN-07	Networks shall be exportable/importable in XMLBIF, and should be importable from GeNIe (.xdsl).	S	T
BBN-08	Bowtie threats and barriers, and FTA structures, may be converted into a BBN skeleton for further editing.	C	D

4.15 Reporting and dashboards (REP)

ID	Requirement	Pri.	Verif.
REP-01	The system shall generate a Safety Assessment Report (DOCX and PDF) following the structure of Manual §18.3, including selected worksheets, diagrams and calculations, from an administrable template.	M	T
REP-02	Every generated report shall state the assessment version, the calculation-engine versions and the risk-scheme version used.	M	I
REP-03	A dashboard shall show: hazards by residual risk region and unit; overdue actions and reviews; barrier health (effectiveness and linked SPI status) for the bowtie library; assessments by status.	M	D
REP-04	All tables shall be exportable to XLSX/CSV and all diagrams to SVG/PNG.	M	T

4.16 Administration (ADM)

ID	Requirement	Pri.	Verif.
ADM-01	Administrators shall configure the risk classification scheme: severity and likelihood levels, labels (ID/EN), quantitative bands, matrix regions, colours and acceptance authorities; changes shall be versioned and never alter locked assessments.	M	T
ADM-02	Administrators shall maintain the organisational structure (directorates, branches, units) used for scoping permissions and reports.	M	T
ADM-03	Method templates (guideword lists, worksheet columns, rating scales) shall be configurable and versioned by Reviewers.	S	T

5. Calculation specifications

The engines below shall be implemented as separately versioned components with documented APIs, so they can be tested independently of the user interface.

5.1 Three-process model of alertness

Time t is in hours, simulated on a fixed step Δt (default 5 min). At each step the state is sleep or wake. Let S be homeostatic level:

Quantity	Update / formula	Default parameter
S while awake	$S \leftarrow L_a + (S - L_a) \cdot e^{d \cdot \Delta t}$	$L_a = 2.4, d = -0.0353$
S while asleep	$S \leftarrow H_a - (H_a - S) \cdot e^{g \cdot \Delta t}$; optional brake: above $S = 12.2$ recovery follows the braked form of Åkerstedt & Folkard	$H_a = 14.3, g = -0.381$, brake off by default
C	$C(t) = C_a \cdot \cos(\pi(t - p)/12)$, t as clock time	$C_a = 2.5, p = 16.8$
W (awake only)	$W = W_c \cdot e^{W_d \cdot \tau}$, τ = hours since last wake	$W_c = -5.72, W_d = -1.51$
U (optional)	12-hour ultradian component	Off by default
Alertness	$A = S + C + W (+ U)$, reported only during wake	—
KSS	$KSS = a + b \cdot A$	$a = 10.6, b = -0.6$
Initial condition	S at simulation start (user-set or spin-up of ≥ 3 days)	$S_0 = 13.0$

5.2 Fault tree quantification

- Minimal cut sets by MOCUS or BDD; results must be identical for trees without truncation.
- Top-event probability: exact via BDD; rare-event approximation $\sum P(\text{cut set})$; min-cut upper bound $1 - \prod(1 - P(\text{cut set}))$.
- Rate conversion: $P = 1 - e^{-\lambda t}$; unavailability of a repairable component $Q \approx \lambda \cdot \text{MTTR} / (1 + \lambda \cdot \text{MTTR})$; periodically tested standby component $Q \approx \lambda \cdot T / 2$.
- Birnbaum $I_B(e) = P(\text{top} | e = 1) - P(\text{top} | e = 0)$; Fussell-Vesely $I_{FV}(e) = [P(\text{top}) - P(\text{top} | e = 0)] / P(\text{top})$.
- β -factor CCF: for a group with rate λ , independent part $(1 - \beta)\lambda$ per member and one common event $\beta\lambda$.

5.3 LOPA, FMECA and risk mapping

- LOPA: $f_C = f_{IE} \times \text{PP}_{CM} \times \text{PPFD}_i$; required additional risk reduction = $\max(1, f_C / f_{\text{target}})$.
- FMEA RPN = $S \times O \times D$; FMECA $C_m = \beta \cdot \alpha \cdot \lambda_p \cdot t$; item criticality $C_r = \sum C_m$ for modes in a severity class.
- Frequency-to-likelihood mapping uses the configured bands with inclusive lower and exclusive upper bounds, after checking unit compatibility.

5.4 Bayesian inference

Exact inference by variable elimination or junction tree; results must agree with brute-force enumeration of the joint distribution to within $1e-9$ for networks of up to 12 binary nodes (used as a test oracle).

6. Non-functional requirements

6.1 Performance and capacity (PER)

ID	Requirement	Pri.	Verif.
PER-01	Interactive pages and worksheets shall load within 2 s (95th percentile) for 100 concurrent users on the reference environment.	M	T
PER-02	Diagram editors shall remain responsive (under 200 ms per edit) for bowties of 60 barriers, fault trees of 500 events, FRAM models of 60 functions and BBNs of 60 nodes.	M	T
PER-03	Minimal cut sets for a 2 000-event tree shall be computed within 60 s; a fatigue simulation of one roster (28 days, 50 staff) within 10 s.	S	T
PER-04	The system shall hold at least 10 years of assessments (estimate 500 assessments and 20 000 hazard-log objects per year) without degradation.	M	A

6.2 Security (SEC)

ID	Requirement	Pri.	Verif.
SEC-01	Authentication shall use AirNav single sign-on (SAML 2.0 or OpenID Connect) with multi-factor authentication; local accounts only for break-glass administration.	M	T
SEC-02	Authorisation shall be role-based and scoped by organisational unit and confidentiality marking; the Administrator role shall not grant approval rights.	M	T
SEC-03	All traffic shall use TLS 1.2 or higher; data at rest (database, object store, backups) shall be encrypted with AES-256 or equivalent.	M	I
SEC-04	The application shall meet OWASP ASVS Level 2 and pass an independent penetration test before go-live and after major releases.	M	T
SEC-05	An append-only audit trail shall record creation, change (old/new value), approval, export and deletion of every object, with user and timestamp, retained for at least 10 years.	M	T

6.3 Privacy (PRV)

ID	Requirement	Pri.	Verif.
PRV-01	Personal data (participants, roster and sleep data, fatigue reports) shall be processed in line with [S4], with a documented purpose and retention period per data category.	M	I
PRV-02	Individual fatigue data shall be pseudonymised in all views except for the Fatigue analyst role, and exports of identifiable fatigue data shall require a recorded justification.	M	T
PRV-03	Participants in a study may be recorded by role instead of name where a just-culture policy requires it.	S	T

6.4 Availability, backup and recovery (AVL)

ID	Requirement	Pri.	Verif.
AVL-01	Availability shall be at least 99.5% during 06:00–22:00 WIB on working days, measured monthly, excluding planned maintenance announced 48 h ahead.	M	A
AVL-02	Backups shall allow recovery with a recovery point objective of 24 h and a recovery time objective of 8 h; restore shall be tested at least twice a year.	M	T

6.5 Usability, language and accessibility (USA)

ID	Requirement	Pri.	Verif.
USA-01	All UI text, help and generated report headings shall be available in Bahasa Indonesia and English, switchable per user.	M	I
USA-02	Each method module shall provide contextual help that links to the corresponding Manual chapter and shows the method's worked example.	S	D
USA-03	The UI shall conform to WCAG 2.2 level AA; colour shall never be the only carrier of meaning (risk regions also carry text labels).	S	T
USA-04	A new assessor with the Manual training shall complete a HAZID study and hazard-log entry in the tool without assistance within 30 minutes (usability test with at least five users).	S	T

6.6 Calculation assurance and maintainability (QA)

ID	Requirement	Pri.	Verif.
QA-01	Each calculation engine shall have a version number, a written specification (Section 5), unit tests with at least 90% branch coverage, and the reference test cases of Section 9 in its automated test suite.	M	I
QA-02	Engine verification shall be performed by a person or team independent of the engine's developer, and recorded before each release that changes an engine.	M	I
QA-03	Results stored with an assessment shall record the engine version; re-running with a newer engine shall create a new result, not overwrite the old one.	M	T
QA-04	Source code, build scripts and documentation shall be delivered to AirNav (or escrowed), with an OpenAPI description of all services.	M	I

7. External interfaces

Interface	Direction	Content	Mechanism	Pri.
Identity provider	In	Users, groups, MFA	SAML 2.0 / OIDC; SCIM for provisioning	M
Occurrence reporting / investigation system	In	Occurrence ID, date, unit, classification, barriers failed	REST API or scheduled file exchange	S
Rostering system	In	Planned and actual duty periods per staff member	REST API or CSV	M (for FAT)
CNS/ATM maintenance management	In	Equipment list, failure records, MTBF/MTTR	REST API or XLSX	S
Document management	Out	Approved Safety Assessment Reports	API upload with metadata	S
E-mail / notifications	Out	Assignments, reminders, approvals	SMTP; optional chat webhook	M
Public API	In/Out	All core objects	REST/JSON, OpenAPI 3, token-based	S

Import/export formats: XLSX/CSV (worksheets, hazard log), DOCX/PDF (reports), SVG/PNG (diagrams), Open-PSA MEF (fault trees), XMLBIF (BBN), FRAM Model Visualiser files (FRAM), documented JSON for bowties and full assessment archives.

8. Delivery phases

Phase	Content	Rationale
1 — Core (MVP)	COM, HZL, REP (basic), ADM, SEC, PRV; methods HAZID, HAZOP, JHA, Bowtie, FMEA (RPN)	Replaces the spreadsheets most used today and establishes the single hazard log and risk scheme.
2 — Quantitative	FHA, FTA engine, LOPA, FMECA criticality, frequency-to-likelihood mapping, dashboards	Supports system changes needing safety objectives and quantitative arguments.
3 — Systems and human performance	STPA, Fatigue model with roster integration, FRAM, BBN, occurrence integration	Adds methods for automation, complex socio-technical work and FRMS.

Each phase ends with acceptance testing against the requirements it delivers and the reference cases in Section 9 relevant to it.

9. Acceptance and reference test cases

The engines shall reproduce the worked examples of the Manual. Inputs are exactly those printed in the Manual; tolerances are relative unless stated.

Test	Input (Manual reference)	Expected result	Tolerance
TC-RISK-01	Severity B, likelihood 3 (Manual Ch. 3)	Index 3B; region Tolerable (upper); authority Accountable Executive	Exact
TC-RISK-02	Severity A, likelihood 3	Index 3A; Intolerable; approval blocked	Exact
TC-FMEA-01	Receiver degradation S=6, O=3, D=7 (Manual §10.3)	RPN 126; ranked first	Exact
TC-LOPA-01	f_{IE} 0.5/yr; P_{CM} 0.05; three IPLs PFD 0.1 (Manual §11.3)	$f_C = 2.5 \times 10^{-5}$ /yr; ratio to 1×10^{-5} target = 2.5	1e-9 abs.
TC-LOPA-02	As TC-LOPA-01 plus IPL PFD 0.1	2.5×10^{-6} /yr; target met	1e-9 abs.
TC-FTA-01	Tree of Manual Figure 7, probabilities as printed	Minimal cut sets {E4}, {E5,E6}, {E7,E8,E9}, {E1,E2,E3}; top = 1.2200×10^{-6}	0.1%
TC-FTA-02	Same tree	Fussell-Vesely: E4 0.820, E5 0.164, E7 0.0164; Birnbaum E5 0.0100	0.5%
TC-BBN-01	Network and CPTs of Manual §17.3	$P(E) = 0.00446$; $P(L) = 2.453 \times 10^{-4}$	1e-6 rel.
TC-BBN-02	Evidence L = yes	$P(F = \text{yes}) = 0.3901$; $P(W = \text{high}) = 0.5448$; $P(S = \text{no}) = 0.1091$	1e-4 abs.
TC-BBN-03	Evidence F = yes; then F = yes and W = high	$P(L) = 6.38 \times 10^{-4}$; 1.10×10^{-3}	1e-6 rel.
TC-FAT-01	Manual §15.4 no-nap roster; defaults of §5.1; $\Delta t = 5$ min; $S_0 = 13.0$ at 00:00 day 1 (asleep); sleeps 23:00–07:00 (days 1–2), 07:00–13:00 day 3	Night-duty minimum A = 5.18 at 05:55; max KSS 7.49; 39.6% of duty at $KSS \geq 7$	± 0.05 (A), ± 5 min, ± 1 pt
TC-FAT-02	As TC-FAT-01 plus nap 15:00–16:30 day 2	Minimum A = 6.30; max KSS 6.82; 0% at $KSS \geq 7$	As above
TC-FHA-01	Severity B, target region Tolerable (lower) (Manual §12.3)	Objective $\leq 1 \times 10^{-7}$ per operating hour	Exact

Verification of reference values. Values for TC-FAT-01/02 were computed with the simplified model (no brake, no ultradian term) and exact exponential updates per step, as specified in §5.1. The supplier shall confirm them independently before they are used for acceptance.

Appendix A. Traceability to the Manual

Manual chapter	Requirement groups
2 SRM process; 5 Workshops	COM-01 to COM-16
3 Risk classification scheme	COM-07 to COM-10, ADM-01, TC-RISK
4 Method selection	COM-03
6 Bowtie	BOW-01 to BOW-08
7 HAZID	HZD-01 to HZD-04
8 HAZOP	HZP-01 to HZP-04
9 JHA	JHA-01 to JHA-04
10 FMEA/FMECA	FME-01 to FME-05, TC-FMEA
11 LOPA	LOP-01 to LOP-05, TC-LOPA
12 FHA	FHA-01 to FHA-04, TC-FHA
13 STPA	STP-01 to STP-07
14 FTA	FTA-01 to FTA-08, §5.2, TC-FTA
15 Fatigue modelling	FAT-01 to FAT-08, §5.1, PRV, TC-FAT
16 FRAM	FRM-01 to FRM-05
17 BBN	BBN-01 to BBN-08, §5.4, TC-BBN
18 Integration, hazard log, report	HZL-01 to HZL-05, REP-01 to REP-04
Technology stack (this SRS)	Appendix C

Appendix B. Glossary

Term	Meaning
Assessment	A documented safety assessment of one change or question, containing one or more method studies.
Method study	One application of one method (e.g. a HAZOP of the cut-over plan) within an assessment.
BDD	Binary decision diagram — data structure for exact Boolean function evaluation, used for fault-tree quantification.
MOCUS	Method for obtaining cut sets — classic top-down algorithm for minimal cut sets.
Open-PSA MEF	Open-PSA Model Exchange Format, an XML format for fault trees and event trees.
XMLBIF	XML Bayesian Interchange Format for Bayesian networks.
MoSCoW	Must / Should / Could (/ Won't) prioritisation.
Other terms	See the Manual, Section 1.3 and Appendix A.

Appendix C. Recommended technology stack

This appendix records the recommended technology stack for ARAP and how it compares with BowTieXP, the most widely used commercial bowtie tool, which is used here as a benchmark. It is a recommendation, not a requirement: a supplier may propose equivalents that meet Sections 4–7. The reference implementation published with this SRS (repository *ARAP---AirNav-Risk-Analysis-Platform*) follows this stack.

C.1 Benchmark: BowTieXP

BowTieXP (now part of the Wolters Kluwer Enablon portfolio as the “BowTieXP Suite”) began as a desktop editor and is now also offered as a browser-based enterprise platform with a central risk repository, barrier-performance monitoring, dashboards, multi-user collaboration, integrations, audit workflows and an incident module (IncidentXP). The capabilities ARAP must match or exceed are summarised below.

BowTieXP capability	How ARAP’s stack matches it
Bowtie editor with strong automatic layout (threats left, consequences right, barriers ordered on each path)	React Flow canvas with a purpose-built, deterministic bowtie layout routine; ELK.js for free-form diagrams (fault trees, STPA, BBN)
Rich barrier metadata and colour-coding (effectiveness, owner, type); barriers shared across paths	Custom React node components; a barrier is stored once and referenced on every path
Central repository, dashboards and barrier health	PostgreSQL as single source of truth; ECharts dashboards in the application; Apache Superset/Metabase on a reporting replica
Incident analysis linked to failed barriers	Occurrence-to-barrier links; barrier-failure counts shown on the bowtie
High-quality printing and export	Server-side SVG/PDF rendering with headless Chromium (Playwright); DOCX via templates
Scope: bowtie and incident analysis	ARAP extends to all twelve methods with one hazard log and one risk scheme

C.2 Recommended stack

Layer	Recommendation	Rationale / notes
Front end	React + TypeScript (Vite); Ant Design or MUI components; TanStack Query; Zustand; react-i18next (ID/EN)	Largest ecosystem and hiring pool; all serious diagram libraries support React
Diagrams	React Flow (@xyflow/react, MIT) + ELK.js; custom bowtie layout. Commercial alternatives: yFiles for HTML, JointJS+, GoJS	One engine for bowtie, FTA, STPA, FRAM and BBN. Prototype the bowtie editor with both free and commercial options before committing
Worksheets	AG Grid (Community; Enterprise if advanced features are needed)	Spreadsheet-like editing for HAZID, HAZOP, JHA, FMEA
Live workshops	WebSockets with record locking and presence; Yjs (CRDT) for true co-editing later	Keeps release 1 simple
Back end	Python 3.12 + FastAPI, modular monolith; Pydantic; SQLAlchemy 2 + Alembic; background jobs with Dramatiq or Celery + Redis	Calculation engines are natural in Python; OpenAPI generated automatically and typed client generated for the front end
Calculation engines	FTA: in-house BDD engine (SCRAM, GPL-3.0, as a test oracle only); BBN: exact variable elimination (in-house or pgmpy, MIT); fatigue, LOPA, FMECA, Monte Carlo: NumPy/SciPy	Versioned and independently verified (QA-01 to QA-03); reference cases of Section 9 in CI
Identity	Keycloak (OIDC/SAML, MFA, LDAP/AD federation), self-hosted	Meets SEC-01; can be hosted in Indonesia
Database	PostgreSQL 16+: relational core; JSONB for method models; recursive CTEs for trees; pg_trgm for duplicate detection and search; row-level security; trigger-based audit tables	No separate graph database needed: models have hundreds, not millions, of nodes
Files and cache	MinIO (S3-compatible) for attachments and reports; Redis for cache and job queue	Self-hostable

Layer	Recommendation	Rationale / notes
Reporting	docxtempl / python-docx for Word; Playwright (headless Chromium) for PDF	Diagrams print exactly as shown on screen
Analytics	Apache ECharts in the application; Apache Superset or Metabase (or Power BI if already standard) on a read-only reporting replica; JupyterHub for analysts	Operational dashboards in-app, organisation-wide BI outside the transactional database
Platform	Docker; Kubernetes (Docker Compose initially); OpenTelemetry, Prometheus/Grafana, Sentry; pytest, Vitest, Playwright; Semgrep, OWASP ZAP, dependency scanning	Meets AVL, QA and SEC groups

C.3 Alternative for .NET or Java organisations

If AirNav IT standardises on .NET or Java, keep the same front end and PostgreSQL, build the core services in ASP.NET Core or Spring Boot, and run the calculation engines as a separate Python service behind an internal API. This adds one service to operate but lets the existing team maintain the core.

C.4 Early decisions

1. **Free versus commercial diagram library.** React Flow is free and flexible but the bowtie layout, printing and undo must be built; yFiles or JointJS+ reduce that effort for a licence fee. The bowtie editor is the largest front-end item, so a two- to three-week prototype with both options is recommended before committing.
2. **Build versus hybrid.** A hybrid option is to license BowTieXP for bowtie work and build ARAP for the other eleven methods, importing BowTieXP XML exports into the ARAP hazard log. This is faster but means two licences and bowtie data held in two places.